

Fractional covering spectra and stability of intersecting extremizers

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We determine the sharp asymptotic fractional vertex-cover frontier at every finite edge/rank ratio, characterize the near-extremizers at its integer design boundaries, and determine the entire frontier with any fixed matching bound. A finite allocation formula also identifies the limit when the matching number diverges. The finite upper bounds require no linearity or small-codegree assumption. The near-design statement requires uniform edges; that hypothesis is retained throughout Part D. These are fractional assertions except for the explicitly designated integer-cover corollaries.

The paper has three connected parts, each with a complete proof. Definitions, theorems, equations and sections are prefixed F, D or S. References within a part use its own prefix; the common finite LP duality argument is included rather than left to an external computational solver. The repeated local conventions make the changes in hypotheses visible. Part S needs an anchored inequality, not a partition into intersecting subfamilies: arbitrary pairs of edges in an anchor group may be disjoint. Its common-rank constructions work at every sufficiently large integer rank, which is essential for both sharpness claims.

Classical inputs are Wilson's fixed-block design existence theorem and Kahn's small-intersection covering corollary, stated precisely in the proof. They enter the sharpness constructions and designated integer-cover corollaries, not the finite signed-counting upper bounds. Füredi–Kahn–Seymour is prior context and is not claimed to be resolved in its general weighted nonuniform formulation. The results neither settle Ryser's conjecture nor Kahn's triple-intersection question. An exact finite maximum for every arithmetic pair of rank and edge count is also outside the statements.

The accompanying reproduction guide pins the three source papers and their checkers. Lean establishes the indicated finite incidence/anchor/extraction lemmas, not Wilson's theorem, the LP duality appendix, all constructions, or the asymptotic spectrum. The written proofs supply these remaining steps. Literature comparison is provisional; no historical priority claim is made.

Part F: The intersecting frontier

F.1. Definitions and the finite bound

Let H be a finite simple intersecting hypergraph with $m \geq 1$ nonempty edges of size at most r , where $r \geq 2$ is an integer. Its fractional vertex cover number $\tau^*(H)$ is the minimum sum of nonnegative vertex weights whose sum on every edge is at least one. By finite LP duality, it equals the maximum $Y = \sum_E y_E$ over weights satisfying

$$y_E \geq 0, \quad \sum_{E \ni v} y_E \leq 1 \quad \text{for every vertex } v. \quad (\text{F.1})$$

Appendix F.A specifies the finite duality fact used. Every edge is nonempty, so feasible edge weights lie in $[0,1]$ and an optimum exists.

Theorem F.1 (finite bound, without additional intersection restrictions). For every integer $k \geq 2$,

$$\tau^*(H) \leq \max \left\{ \frac{(k-1)r+1}{k}, \frac{m}{k+1} \right\}. \quad (\text{F.2})$$

For $m \geq 2$ we also have $\tau^*(H) \leq m/2$; for $m = 1$ the value is one. No partite, linearity, uniformity, or small-codegree condition is needed. The finite maximum in (F.2) is not asserted to be the exact optimum for every arithmetic pair (r, m) .

F.2. A signed bin inequality

Lemma F.2. Suppose $k \geq 2$, $1/(k+1) < b \leq 1/k$, and a bin contains $\ell \in \mathbb{Z}_{\geq 0}$ weights in $[0, b]$ of total $W \leq 1 - b$. Then

$$W - (1 - kb)\ell \leq (k-1)\{(k+1)b - 1\}. \quad (\text{F.3})$$

This is an inequality about the weights assigned to the bin, not all edges incident at the bin's vertex.

Proof. Put $p = (k+1)b - 1 > 0$ and $B = 1 - kb \geq 0$. If $\ell \leq k-1$, use $W \leq \ell b$ to get

$$W - B\ell \leq \ell\{(k+1)b - 1\} \leq (k-1)p.$$

If $\ell \geq k$, use $W \leq 1 - b$ and $B \geq 0$ to get

$$W - B\ell \leq 1 - b - kb = (k^2 - 1)b - (k-1) = (k-1)p.$$

The empty bin belongs to the first case. This proves the lemma. The signed cost is essential: individual bin scores $W - B\ell$ need not be positive. We will sum them only over a genuine partition.

F.3. Proof of the finite theorem

Fix any feasible vector (F.1), choose a maximum-weight edge P , and write $b = y_P$, $Y = \sum_E y_E$. Choose any v in P ; feasibility gives $b \leq 1$. Nonnegative weights and intersectingness imply the weighted-star inequality

$$Y - b \leq \sum_{v \in P} \sum_{E \neq P, v \in E} y_E \leq |P|(1 - b) \leq r(1 - b). \quad Y \leq r - (r-1)b. \quad (\text{F.4})$$

Multiple intersections do not invalidate this inequality.

Fix k and put $g = (k-1)r + 1$. If $b \leq 1/(k+1)$, then $Y \leq m/(k+1)$, proving (F.2). If $b > 1/k$, (F.4) gives $Y \leq r - (r-1)/k = g/k$, again proving (F.2).

It remains to consider $1/(k+1) < b \leq 1/k$. For every edge E other than P , choose **one** vertex $v(E) \in E \cap P$. This is possible by intersectingness. Assign its weight y_E to that vertex's bin. Each edge is assigned exactly once, even if it meets P repeatedly. At a vertex v of P the assigned weights are a subset of the other incident edge weights; therefore their total W_v is at most $1-b$. Each individual weight is at most b . Let ℓ_v count the assigned edges. We have the exact partition identities

$$\sum_{v \in P} W_v = Y - b, \quad \sum_{v \in P} \ell_v = m - 1. \quad (\text{F.5})$$

Lemma F.2 can now be summed, even though its summands have signs:

$$(Y - b) - (1 - kb)(m - 1) \leq |P|(k-1)\{(k+1)b - 1\} \leq r(k-1)\{(k+1)b - 1\}. \quad (\text{F.6})$$

The last factor is positive. Rearranging yields

$$Y \leq m - g + b\{(k+1)g - km\}. \quad (\text{F.7})$$

The right side is affine in b . At the endpoints $b=1/(k+1)$ and $b=1/k$ its values are respectively $m/(k+1)$ and g/k . Therefore (F.7) is at most their maximum throughout the closed interval. This proves (F.2) for every feasible vector and hence for the optimum.

For $m \geq 2$, each pair of distinct edges has a common vertex, so $y_E + y_F \leq 1$. Summing over all unordered edge pairs gives $(m-1)Y \leq \binom{m}{2}$, hence $Y \leq m/2$. For $m = 1$, a unit vertex weight and a unit dual edge weight show $\tau^* = 1$.

This upper-bound proof uses no design theorem, rounding result, numerical experiment or claim in OpenAI/math.

F.4. The exact limiting curve

Define the continuous nondecreasing function

$$\psi(c) = c/2 \quad (0 \leq c \leq 1), \quad \psi(c) = \max \left\{ \frac{a}{a+1}, \frac{c}{a+2} \right\} \quad (a \leq c \leq a+1, \ a = 1, 2, \dots). \quad (\text{F.8})$$

The formulas agree at integer endpoints. On $[a, a+1]$ the switch is $c = a(a+2)/(a+1) = a+1-1/(a+1)$: a plateau is followed by a linear ramp. In particular $\psi(3/2) = 1/2$, $\psi(7/4) = 7/12$, and $\psi(5/2) = 2/3$.

Theorem F.3 (sharp frontier at every real ratio). If r_j tends to infinity, H_j is as in Theorem F.1, and $m_j/r_j \rightarrow c < \infty$, then

$$\limsup_j \tau^*(H_j)/r_j \leq \psi(c). \quad (\text{F.9})$$

For every real $c \geq 0$ there is a sequence of **simple intersecting r_j -uniform** hypergraphs with $m_j/r_j \rightarrow c$ for which

$$\tau^*(H_j)/r_j \rightarrow \psi(c). \quad (\text{F.10})$$

Thus the same sharp limiting curve holds whether uniformity is required or only the rank bound is imposed.

Upper bound. For $c \geq 1$ choose the fixed $k = a+1$ corresponding to an interval $[a, a+1]$ containing c , and apply (F.2). Dividing by r_j and taking limsup gives (F.9). At shared endpoints either formula works. For $c \leq 1$ use the pair bound, with the single-edge cases contributing at most $1/r_j$. This also handles $c = 0$.

The next section proves (F.10) and states every imported theorem.

F.5. Sharpness and classical inputs

A Steiner 2-($v, k, 1$) design is a family of k -point blocks on a v -point set, such that every pair of distinct points belongs to exactly one block. Each point belongs to $r_0 = (v-1)/(k-1)$ blocks. For replication at least two, its incidence dual has v distinct edges of size r_0 and is simple and intersecting, and every active vertex has degree k . Uniform dual weights $1/k$ and uniform primal weights $1/r_0$ give matching values v/k , since there are vr_0/k blocks. Thus its fractional cover number is exactly v/k .

Input W (Wilson 1975, fixed-block design existence). For every fixed $k \geq 2$ and all sufficiently large v satisfying

$$k-1 \mid v-1, \quad k(k-1) \mid v(v-1), \quad (\text{F.11})$$

a Steiner 2-($v, k, 1$) design exists. In particular one can take an unbounded sequence $v = 1 + k(k-1)N$, with replication $r_0 = kN$. This classical existence theorem is used only for sharpness, not for the finite upper bound.

Input K (Kahn 1994, Corollary F.5.4, printed p.140). For fixed $C > 0$, intersecting r -uniform families with at most Cr edges and maximum distinct-edge intersection $o(r)$ have integer cover number at most $(C/(C+1) + o(1))r$. Again this is used only in a construction step.

F.5.1 Plateau construction

Fix $a \geq 1$ and $k = a + 1$. Apply Input W and let H_0 be a design dual with $r_0 \rightarrow \infty$ and $v = (k-1)r_0 + 1$. It is linear, with $v/r_0 \rightarrow a$. Applying Input K first with $C = a + \eta$ and then decreasing η to zero gives an integer cover C_0 of size

$$|C_0| \leq (a/(a+1) + o(1))r_0 < r_0 \quad (\text{F.12})$$

for all sufficiently large r_0 . C_0 is nonempty. All its vertices are original design blocks. The number of original vertices is $n_0 = vr_0/k = \Theta(r_0^2)$.

For any fixed $c \geq a$ choose $L = \max\{0, \lfloor cr_0 \rfloor - v\}$. This is $O(r_0)$. Enlarge C_0 to a fixed set T of $r_0 - 1$ original vertices. For each of L distinct original vertices p outside T , add the edge $T \cup \{p\}$, avoiding edges already in H_0 . There are $n_0 - r_0 + 1$ candidates and at most v forbidden originals, so enough remain. The new edges are distinct r_0 -sets; they meet one another through T and meet every original edge through C_0 . Thus the enlarged hypergraph is simple and intersecting.

Put dual weight $1/k$ on the old edges and zero on the new ones. It remains feasible. Put primal weight $1/r_0$ on every original vertex; every edge, including every new one, has exactly r_0 original vertices, so its primal sum is one. Both objectives are v/k . Therefore the enlarged family has exactly

$$\tau^* = v/k, \quad m = v + L, m/r_0 \rightarrow c, \quad \tau^*/r_0 \rightarrow a/(a+1). \quad (\text{F.13})$$

No explicit algorithm to find Wilson's designs or C_0 is asserted; their existence follows from the two named published inputs. For prime-power k , affine-line designs give explicit design instances.

F.5.2 Linear-ramp construction

For c in $(0, 1]$ use the complete pair design on v points (block size two). For c in $[a, a+1]$, $a \geq 1$, use Input W with block size $a+2$. Let r_0 be its replication and set $r = \lceil v/c \rceil$. In both cases $r \geq r_0$. Pad each original edge with $r - r_0$ distinct private vertices, different across edges. Uniformity becomes r ; all original pair intersections remain exactly one, so simplicity and intersectingness are preserved.

Keep uniform dual weights $1/k$, where k is the block size, and primal weights $1/r_0$ on old vertices and zero on the padding. These are feasible with equal value v/k . Hence

$$m/r \rightarrow c, \quad \tau^*/r \rightarrow c/k. \quad (\text{F.14})$$

This gives $c/2$ below one and $c/(a+2)$ on $[a, a+1]$. Choose between (F.13) and (F.14) according to the larger term in (F.8), proving sharpness. For $c = 0$ take one r -edge with $r \rightarrow \infty$; its value one divided by r tends to zero. Theorem F.3 is proved.

F.5.3 Explicit partite sharpness by shifting a partial pencil

The preceding all-integer construction inputs are unnecessary when the block size is a prime power. We now give an explicit construction that additionally preserves a partition into as many parts as the rank.

Proposition F.3.1. Every plateau for which $a + 1$ is a prime power, and every ramp for which $a + 2$ is a prime power, is attained by simple intersecting partite uniform families with **equal integer and fractional cover numbers**. In particular the entire curve (F.8) is sharp within partite uniform families for $0 \leq c \leq 4$.

Proof. Fix a prime power q , let $V = \mathbb{F}_q^N$ with $N \geq 3$, and put $v = q^N$, $r_0 = (v - 1)/(q - 1)$. Vertices of the hypergraph are affine lines; its original edge at $p \in V$ is the set of all lines through p . For each one-dimensional linear direction D , the lines of that direction form one part. There are r_0 parts and each original edge uses exactly one vertex of each part. Any two original edges meet at their unique common line. Every line belongs to q original edges. Uniform old-edge weights $1/q$ have value v/q . All v/q vertices in a fixed part form an integer cover with the same value.

For each two-dimensional linear subspace W choose $z_W \in W \setminus \{0\}$. Define a new edge Q_W by choosing, for each direction D , the line

$$\begin{cases} z_W + D, & D \subset W, \\ D, & D \not\subset W. \end{cases} \quad (\text{F.14a})$$

This still uses one old vertex per part. These lines cover all points of V : a point in W lies on a line through z_W of a direction inside W ; a point outside W lies on its line through zero of a direction outside W . Thus Q_W meets every original edge. Two such new edges share the zero-line in any direction outside both subspaces. Such a direction exists because

$$r_0 > 2(q + 1) \quad (N \geq 3, q \geq 2).$$

Indeed r_0 is at least $q^2 + q + 1$, and $q^2 - q - 1$ is positive. Therefore all new edges are pairwise intersecting.

The new edge changes precisely the q directions in W other than $\langle z_W \rangle$. At least two changed directions span W ; hence different subspaces give different new edges. Each new edge also retains at least two distinct lines through zero, since $r_0 - (q + 1) \geq q^2$. If it were an original point-pencil, that point would be zero, contradicting the changed directions. Thus all old and new edges are distinct.

The number of two-dimensional linear subspaces is

$$\frac{(v - 1)(v - q)}{(q^2 - 1)(q^2 - q)} = \Theta(v^2). \quad (\text{F.14b})$$

This follows by counting ordered independent vector pairs and then ordered bases of a fixed two-dimensional space. For any fixed $c \geq q - 1$, add $L = \max\{0, \lfloor cr_0 \rfloor - v\}$ different new edges. There are more than enough choices since $L = O(r_0)$. Extend the old dual weights by zero. A whole original part remains an integer cover for every new edge, so

$$\tau(H) = \tau^*(H) = v/q, \quad m/r_0 \rightarrow c, \quad \tau^*(H)/r_0 \rightarrow (q - 1)/q. \quad (\text{F.14c})$$

Here τ denotes the ordinary minimum number of vertices meeting every edge. This proves the claimed plateau sharpness.

For a ramp, use just the original affine-line design dual and pad its edges privately to rank $r = \lceil v/c \rceil$, where $0 < c \leq q - 1$. Put the private vertex in position j of every edge into its own new part j , distinct across edges. There is exactly one vertex in each old or new part on every edge. One old part is still an integer cover, and the old dual weights remain feasible. Thus $\tau = \tau^* = v/q$ and $\tau^*/r \rightarrow c/q$. For $q=2$ this covers the interval below one; for $q=a+2$ it gives the ramp in (F.8). The adjacent block-size pairs (2,3), (3,4) and (4,5) cover all intervals up to $c=4$. At $c=0$ a single edge with singleton parts suffices. This proves the final claim. Finite fields of prime-power order are the standard algebraic input; Wilson and Kahn are not used in this explicit proposition.

These plateau examples permit very large repeated intersections. If $c > q - 1$, the new edges alone give

$$I(H) = \sum_{\{E,F\}} (|E \cap F| - 1) \geq \binom{L}{2} \{r_0 - 2(q+1) - 1\},$$

so $I(H)/r_0^2 \rightarrow \infty$. The sharp fractional result therefore covers families outside any small-intersection regime. These examples supply integer lower constructions; they do not give an integer upper bound for arbitrary families.

F.6. Exact equality and deletion stability inside the linear phases

Fix an integer $k \geq 2$, put $g = (k-1)r + 1$, and suppose

$$A = km - (k+1)g > 0. \tag{F.15}$$

Thus the second term in (F.2) is strictly larger than the first.

Theorem F.4 (finite equality). Under (F.15),

$$\tau^*(H) = m/(k+1) \iff \max_v \deg_H(v) \leq k+1. \tag{F.16}$$

Proof. Choose an optimal feasible vector. If its maximum b exceeds $1/k$, the weighted-star bound gives a value at most g/k , strictly smaller than $m/(k+1)$. If $1/(k+1) < b \leq 1/k$, (F.7) rewrites as

$$Y \leq m/(k+1) - A\{b - 1/(k+1)\} < m/(k+1). \tag{F.17}$$

Thus equality requires b at most $1/(k+1)$. Since m such weights sum to $m/(k+1)$, every weight equals $1/(k+1)$. Vertex feasibility implies the degree cap. Conversely that cap makes the uniform edge weights $1/(k+1)$ feasible, attaining the finite upper bound. This proves (F.16).

Theorem F.5 (explicit deletion bound). Under (F.15), write $d = m/(k+1) - \tau^*(H) \geq 0$. If

$$d < A/\{k(k+1)\}, \tag{F.18}$$

there is a set of s edges whose deletion leaves maximum degree at most $k+1$, where

$$s \leq (k+1)(k+2)(1 + m/A)d. \tag{F.19}$$

The number s is an integer, so the displayed real upper bound also bounds it after rounding down.

Proof. Again use an optimal vector with maximum b . Condition (F.18) means $Y > g/k$, so b cannot exceed $1/k$. If b exceeds $1/(k+1)$, (F.17) implies

$$b - 1/(k+1) \leq d/A. \tag{F.20}$$

Delete precisely the edges with $y_E \leq 1/(k+2)$. Each retained edge has weight strictly above that threshold, so feasibility allows at most $k+1$ retained edges at any vertex. If b is at most $1/(k+1)$, the nonnegative deficits $1/(k+1) - y_E$ sum to d . Each deleted edge contributes at least $1/\{(k+1)(k+2)\}$; hence s is at most $(k+1)(k+2)d$. If b exceeds $1/(k+1)$, sum the nonnegative peak deficits instead:

$$\sum_E (b - y_E) = mb - Y = m\{b - 1/(k+1)\} + d \leq (1 + m/A)d.$$

Each deleted edge again contributes at least $1/\{(k+1)(k+2)\}$. This proves (F.19) in both cases. Empty cores are allowed. No assumption about zero-weight edges is made.

Corollary F.6 (complete stability criterion on a strict ramp). Fix $k \geq 2$ and a real $c \in (k - 1/k, k]$. For any sequence as in Theorem F.3 with $m_j/r_j \rightarrow c$,

$$\frac{\tau^*(H_j)}{r_j} \rightarrow \frac{c}{k+1} \iff \text{deleting } o(r_j) \text{ edges leaves maximum degree at most } k+1. \quad (\text{F.21})$$

For the forward direction, $A_j/r_j \rightarrow kc - k^2 + 1 > 0$, while $d_j/r_j \rightarrow 0$. Therefore (F.18) holds eventually and (F.19) is $o(r_j)$. For the converse, put weight $1/(k+1)$ on retained edges and zero on all deleted edges. This is feasible on the original family and has value $(m_j - o(r_j))/(k+1)$. The finite upper bound and (F.8) complete the squeeze. All parameters k, c remain fixed in this limit.

The phase-switch point $c = k - 1/k$ is excluded: A_j need not be of linear order there, and (F.19) does not give a uniform coefficient. The exclusion is essential, already for $k = 2$. More generally let k be a prime power and take Proposition F.3.1 at $c = k - 1/k$. Then $L/r_0 \rightarrow 1 - 1/k$, and the fractional value is asymptotically sharp. Any $k+2$ new edges retain a common zero-line outside all their selected two-spaces once $r_0 > (k+2)(k+1)$. Therefore any core of maximum degree at most $k+1$ retains at most $k+1$ new edges. It must delete at least $L - (k+1) = \Theta(r_0)$ edges. This disproves the forward implication in (F.21) if the switch is included.

This corollary does not assert integer-cover recovery. Unlike at integer design ratios, the degree cap alone here need not make intersection excess $o(r_j^2)$. A global rounding theorem requires a separate argument. For an explicit illustration, take the $q = 3$ affine design dual and clone every vertex in one parallel class t times, with each copy assigned to its own new part. Every edge now has rank $r = r_0 + t$, every active degree remains three, and a whole old part still gives $\tau = \tau^* = v/3$. Exactly $v/3$ point triples have their pair intersections increased by t , so $I = tv$. If $t/r_0 \rightarrow 2/c - 1$ for fixed $c \in (3/2, 2)$, then $m/r \rightarrow c$, $\tau^*/r \rightarrow c/3 = \psi(c)$, but $I/r^2 \rightarrow c(1 - c/2) > 0$. This verifies why near-linearity of the retained core cannot simply be inferred on a strict noninteger ramp.

F.7. Comparison, verification and remaining questions

This replaces the previous repository's nonlinear fractional envelope ϕ and its conservative noninteger gap by the exact asymptotic frontier ψ . The previous frozen package remains a correct historical proof. At integer ratios the new curve agrees with the known design values; at nonintegers it supplies the complete value rather than an unspecified gap. The separate fractional design-stability theorem and its integer recovery corollary are unchanged.

This is not an integer-cover theorem. No unrestricted Ryser solution, journal acceptance, first-result claim or prize-level assessment is made. The finite optimal value at each arithmetic pair (r, m) and a full classification of noninteger extremizers remain open within this project.

The finite upper bound for a feasible edge-weight vector, including the genuine hypergraph-to-bin assignment and all partition sums, is proved in the pinned Lean project and supported by exact rational replay. Formal coverage is recorded separately; it does not include design existence, published rounding, LP duality or all real-limit arguments. Fixed primal/dual certificates illustrate both finite phases, and no solver output is used as evidence for a theorem. The exploratory LP searches helped select the conjectured curve; the proof above is independent of them.

Appendix F.A. Finite duality

For the finite vertex-by-edge incidence matrix A , weak duality follows by summing the feasibility constraints. A primal vector can be clipped coordinatewise into $[0, 1]$ without destroying feasibility. Every nonempty edge bounds its dual coordinate by one. Thus both extrema exist. Let α be the primal minimum. For $\beta < \alpha$, the point $(\mathbf{1}, \beta)$ lies outside the finitely generated closed cone

$$\{(A^T z - s, \mathbf{1}^T z + t) : z, s, t \geq 0\} \subset \mathbb{R}^{m+1}.$$

Finite-dimensional strict separation gives (u, b) whose value is nonnegative on the cone and negative at this point. The coordinate generators give $u \leq 0$ and $b \geq 0$. Put $y = -u \geq 0$. The vertex generators give $Ay \leq b\mathbf{1}$, while negativity at the separated point gives $\sum_E y_E > b\beta$. If $b = 0$, nonempty edges force $y = 0$, contradicting strict separation. Thus y/b is feasible with value greater than β . Let $\beta \uparrow \alpha$ and use dual compactness. Weak duality then proves equality of the two extrema.

For completeness, the cone is closed as follows. Any representation on a dependent support can lose one generator: move its coefficients along a linear relation until one becomes zero, preserving nonnegativity. Iterate to an independent support. A cone on linearly independent generators is closed because the inverse map on their span is continuous. There are only finitely many such subsets, so the original cone is a finite union of closed cones. The standard finite-dimensional separation theorem is the background input.

References and provenance

[W] R. M. Wilson, *An existence theory for pairwise balanced designs, III: Proof of the existence conjectures*, J. Combin. Theory Ser. A **18** (1975), 71–79, DOI [https://doi.org/10.1016/0097-3165\(75\)90067-9](https://doi.org/10.1016/0097-3165(75)90067-9). The primary abstract states the existence theorem and both divisibility conditions. Its published theorem is an explicitly named construction input.

[K] J. Kahn, *On a Problem of Erdős and Lovász. II: $n(r)=O(r)$* , J. Amer. Math. Soc. **7** (1994), 125–143, Corollary F.5.4, DOI <https://doi.org/10.1090/S0894-0347-1994-1224593-5>. The primary Section F.5 text was read; the integer covering theorem is prior work, not a contribution of this note.

[FKS] Z. Füredi, J. Kahn and P. D. Seymour, *On the fractional matching polytope of a hypergraph*, Combinatorica **13** (1993), 167–180, DOI <https://doi.org/10.1007/BF01303202>. Its rank-dependent weighted matching results are classical context, not a proof input for (F.2). The limited comparison does not certify historical novelty.

Part D: Design-boundary stability

D.1. Conventions and statements

Let H be a finite simple intersecting r -uniform hypergraph, $r \geq 2$, with $m \geq 1$ edges. Vertices outside all edges are ignored. A vertex cover meets every edge, and its minimum size is $\tau(H)$. A fractional

vertex cover has nonnegative vertex weights of sum at least one on each edge; its minimum total is $\tau^*(H)$. Equivalently, by finite LP duality, $\tau^*(H)$ is the maximum of $Y = \sum_E y_E$ over nonnegative edge weights satisfying

$$\sum_{E \ni v} y_E \leq 1 \quad \text{for every vertex } v. \quad (\text{D.1})$$

The optimum exists. Appendix D.A records the duality justification.

For any intersecting subfamily J put

$$I(J) = \sum_{\{E, F\} \subset E(J)} (|E \cap F| - 1). \quad (\text{D.2})$$

Pairs are unordered and distinct. Set $I = 0$ for an empty subfamily. Every summand is nonnegative; $I = 0$ means linearity.

Theorem D.1 (general finite extraction). Fix an integer $k \geq 2$ and any feasible weights (D.1). If $Y > m/(k+1)$, retain exactly the edges with $y_E > 1/(k+1)$. Let J be this core, q its edge count, and $s = m - q$ the number deleted. Then every active vertex of J has degree at most k and

$$s \leq \frac{m(k+1)\{mr - (m+r-1)Y\}}{(r-1)\{(k+1)Y - m\}}. \quad (\text{D.3})$$

Both factors in the denominator are positive. Independently of the mass condition, this same threshold core has maximum degree at most k . It satisfies

$$\sum_v d_J(v)(k - d_J(v)) = q\{(k-1)r - q + 1\} - 2I(J) \geq 0, \quad (\text{D.4})$$

including $q = 0$. In particular $I(J) \leq q\{(k-1)r - q + 1\}/2$.

Theorem D.2 (linear deletion stability at a design boundary). Suppose $m = (k-1)r + 1$, $k \geq 2$, and put

$$d = m/k - \tau^*(H) \geq 0. \quad (\text{D.5})$$

There is a core J , obtained by the threshold above from any optimal dual vector, for which

$$s \leq \frac{2k^2(k+1)r}{r-1} d \leq 4k^2(k+1)d, \quad \max_v d_J(v) \leq k, \quad I(J) \leq qs/2. \quad (\text{D.6})$$

If n counts its active vertices, then

$$\sum_v d_J(v)(k - d_J(v)) = qs - 2I(J), \quad 0 \leq kn - qr \leq qs. \quad (\text{D.7})$$

The dependence on d cannot be replaced by $o(d)$ uniformly, already for $k = 2$; Section D.5 gives a family whose minimum deletion count is $2d$. No optimality of the coefficient in (D.6) is claimed.

Theorem D.3 (complete asymptotic characterization at integer ratios). Fix a positive integer a and a sequence of such hypergraphs with $r_j \rightarrow \infty$ and $m_j/r_j \rightarrow a$. The following are equivalent:

1. $\tau^*(H_j)/r_j \rightarrow a/(a+1)$.
2. Deleting $o(r_j)$ edges leaves a core with maximum vertex degree at most $a+1$.

For the cores in either condition, $q_j/r_j \rightarrow a$,

$$I(J_j) = o(r_j^2), \quad n_j/r_j^2 \rightarrow a/(a+1), \quad \sum_v (d_{J_j}(v) - a - 1)^2 = o(r_j^2). \quad (\text{D.8})$$

Moreover

$$\tau(H_j)/r_j \rightarrow a/(a+1), \quad \tau(H_j)/\tau^*(H_j) \rightarrow 1. \quad (\text{D.9})$$

Only (D.9) uses the published rounding input stated in Section D.4. The hypotheses impose no bound on intersections in the whole family. In fact (D.8) can fail for the whole family even when (D.9) holds exactly, as Section D.5 demonstrates.

D.2. The weighted estimate and finite extraction

Choose an edge P of maximum weight b . Every other edge meets P , so nonnegative weights and (D.1) give

$$Y - b \leq \sum_{v \in P} \sum_{E \neq P, v \in E} y_E \leq r(1 - b). \quad \text{Thus } Y \leq r - (r - 1)b. \quad (\text{D.10})$$

Multiple intersections only increase the middle sum. Since $Y \leq mb$,

$$Y \leq \frac{mr}{m + r - 1}, \quad (r - 1)(mb - Y) \leq mr - (m + r - 1)Y =: \Psi. \quad (\text{D.11})$$

In particular $\Psi \geq 0$. These facts hold for every feasible vector.

For a deleted edge, $b - y_E \geq b - 1/(k + 1)$. Thus

$$s\{b - 1/(k + 1)\} \leq \sum_{E \text{ deleted}} (b - y_E) \leq mb - Y. \quad (\text{D.12})$$

Here all peak deficits $b - y_E$ are nonnegative. Since $b \geq Y/m$ and $Y/m > 1/(k + 1)$, combining (D.11)–(D.12) proves (D.3).

If a vertex belonged to at least $k+1$ retained edges, their weights would sum to more than one. Hence the retained degree is at most k . For the core, double counting incidences and edge-pair intersections gives

$$\sum_v d_J(v) = qr, \quad \sum_v \binom{d_J(v)}{2} = \binom{q}{2} + I(J).$$

Consequently $\sum_v d_J(v)^2 = qr + q(q - 1) + 2I(J)$. Subtracting this from kqr proves (D.4); all summands on its left are nonnegative. This proves Theorem D.1, including empty cores.

D.3. The design boundary and the iff statement

At $m = (k - 1)r + 1$, equation (D.11) gives $\tau^* \leq m/k$, proving $d \geq 0$. For an optimal dual vector, $Y = m/k - d$ and $\Psi = krd$. If $d \leq m/[2k(k + 1)]$, then

$$(k + 1)Y - m = m/k - (k + 1)d \geq m/(2k) > 0.$$

Insert these estimates into (D.3) to obtain the first bound in (D.6). If d is larger instead, the trivial bound $s \leq m$ gives $s \leq 2k(k + 1)d$, which is stronger. This proves (D.6) for every d , with no small-deficit hypothesis. Formula (D.4) becomes the first identity of (D.7). For each active degree $1 \leq d_J(v) \leq k$, $0 \leq k - d_J(v) \leq d_J(v)(k - d_J(v))$. Summing proves the second claim of (D.7). This completes Theorem D.2.

For Theorem D.3 put $k = a + 1$. Under condition 1,

$$\Psi_j/r_j^2 = m_j/r_j - (m_j/r_j + 1 - 1/r_j)Y_j/r_j \longrightarrow 0.$$

Also $Y_j/m_j \rightarrow 1/k > 1/(k+1)$. Applying (D.3) shows $s_j/r_j \rightarrow 0$. This proves condition 2, with an actual threshold core.

Conversely, if a core of size $q_j = m_j - o(r_j)$ has degree at most k , put weight $1/k$ on each core edge and zero on each deleted edge. These weights are feasible for the original family, giving $\tau^*(H_j) \geq q_j/k$. The harmonic upper bound (D.11) proves condition 1. This argument is valid for any such core.

For any core as in condition 2, (D.4) implies

$$0 \leq \sum_v d_J(v)(k - d_J(v)) \leq q\{(k-1)r - q + 1\} = o(r^2), \quad 0 \leq 2I(J) \leq q\{(k-1)r - q + 1\} = o(r^2).$$

The bracket is nonnegative for nonempty cores, by (D.4). Summing $k - d_J$ as above shows $kn - qr = o(r^2)$. Finally, for $1 \leq d_J \leq k$, $(k - d_J)^2 \leq (k-1)d_J(k - d_J)$; summation proves the variance assertion (D.8). This proves every structural part of Theorem D.3 without a rounding theorem.

D.4. Integer rounding and what is imported

Input K (Kahn 1994, Corollary D.5.4, printed p.140). For fixed $C > 0$, an intersecting r -uniform sequence with at most Cr edges and maximum intersection of distinct edges $o(r)$ satisfies

$$\tau/r \leq C/(C+1) + o(1). \quad (\text{D.13})$$

This is a published input, not an assertion proved or formalized in this note. We explain precisely how its hypotheses are reached.

Let J be a core from Theorem D.3 and write $\epsilon = I(J)/r^2 \rightarrow 0$. Put $\rho = \sqrt{\epsilon} + r^{-1/2}$, so $\rho \rightarrow 0$ and $\rho r \geq \sqrt{r}$. Every edge pair with intersection at least ρr contributes at least $\rho r - 1$ to $I(J)$. The number of such pairs is at most

$$I(J)/(\rho r - 1) = o(r).$$

Indeed its ratio to r is $\epsilon/(\rho - 1/r) \rightarrow 0$. Choose and delete one edge from each such pair. At most $o(r)$ edges are deleted, and no bad pair remains. The resulting family J' has maximum distinct-edge intersection $< \rho r = o(r)$.

For each fixed $\eta > 0$, J' has at most $(a + \eta)r$ edges eventually. Applying Input K and then letting η decrease to zero gives $\limsup \tau(J')/r \leq a/(a+1)$. Each deleted edge, from either stage, can be covered with one of its r vertices; thus the total extra cost is $o(r)$. Therefore the same upper bound holds for H . The lower bound $\tau(H) \geq \tau^*(H)$ proves (D.9).

One consequence is a uniform separation statement: for each fixed integer a and $\eta > 0$ there exist $\delta > 0$ and R such that, if $r \geq R$, $|m/r - a| \leq \delta$ and $\tau/r \geq a/(a+1) + \eta$, then $\tau^*/r \leq a/(a+1) - \delta$. If this failed, for $\delta = 1/j$ and $R = j$ one could choose a violating family with $m/r \rightarrow a$, integer cover ratio bounded above the endpoint, and fractional ratio tending to the endpoint by (D.11). This contradicts Theorem D.3. No explicit value of δ , no rate in Input K, and no resolution of unrestricted Ryser's conjecture are asserted.

D.5. Sharp deletion order and unavoidable exceptional edges

Here is a complete elementary family for $k = 2$. Fix even $n \geq 4$ and an integer $t \geq 1$, and put $r = n - 1 + t$. For each pair of distinct indices i, j in $\{1, \dots, n\}$ use a vertex $x_{ij} = x_{ji}$. Define n good edges

$$A_i = \{x_{ij} : j \neq i\} \cup \{u_{i,1}, \dots, u_{i,t}\},$$

where all padding vertices u are distinct and fresh. Thus $|A_i| = r$, and distinct good edges meet exactly once. Let C be the $n/2$ vertices of one fixed perfect matching of the index set. Each good edge contains exactly one vertex of C .

Take a fresh common set W of size $r - n/2 - 1 = n/2 + t - 2$, and fresh distinct vertices $p_1, \dots, p_t$. The t bad edges are

$$B_l = C \cup W \cup \{p_l\}.$$

Each has r vertices; all edges are distinct. Each bad edge meets each good edge exactly once, and two bad edges meet in $r-1$ vertices. The whole family is intersecting, with $m = n + t = r + 1$.

Put dual weight $1/2$ on the good edges and zero on the bad edges. Pair vertices have incident dual weight one, good private vertices have weight $1/2$, and all other vertices have weight zero. This is feasible with total $n/2$. The integer cover C has size $n/2$ and meets every edge. Weak duality therefore certifies

$$\tau = \tau^* = n/2, \quad d = m/2 - n/2 = t/2. \quad (\text{D.14})$$

Deleting all bad edges leaves degree at most two. Conversely, let ℓ bad edges remain in any degree-at-most-two subfamily. At a vertex of C they already have degree ℓ , so $\ell \leq 2$. Each of the $n/2$ disjoint matching pairs originally contributed two good-edge incidences at its vertex; at least ℓ of these two good edges must be deleted. These requirements involve disjoint good-edge sets, so at least $\ell n/2$ good edges are deleted. The total deletion count is consequently at least

$$t - \ell + \ell n/2 \geq t.$$

Its exact minimum is $t = 2d$. This proves the sharp linear order claim.

Only bad–bad pairs contribute excess, so

$$I(H) = \binom{t}{2}(r - 2). \quad (\text{D.15})$$

For the exact integer sequence $n = 2j^4$ and $t = j^3$, $j \geq 2$, we have $t = o(r)$, $m/r \rightarrow 1$ and fractional ratio tending to $1/2$, but $I(H)/r^2 \rightarrow \infty$. Even the integer ratio agrees with its fractional counterpart exactly. Thus Theorem D.3 cannot require the whole original family to have $I = o(r^2)$, nor can it omit deletion.

D.6. Verification scope and significance limits

The finite extraction inequality, threshold degree cap, and uniform-weight converse are formalized for actual finite incidence sets in the accompanying pinned Lean project. The weighted-star proof sums over real edge and vertex sets and uses the genuine uniformity, intersectingness and feasibility hypotheses; these are not replaced by a scalar assumption. Kernel reports contain only `propext`, `Classical.choice` and `Quot.sound`.

The degree/excess double counting, real-sequence limit arguments, LP duality, the sharpness construction and Input K are written proofs, not full-paper Lean formalizations. Exact rational

diagnostics check incidence identities, the extraction bound and explicit primal/dual sharpness certificates; they are not substitutes for the universal proofs above. No external human review, historical priority or high-level prize significance is claimed.

Appendix D.A. The finite duality fact

Let A be the vertex-by-edge incidence matrix. Weak duality follows by summing $\sum_{E \ni v} y_E \leq 1$ against any feasible primal weights. The primal infimum α is attained after clipping each vertex weight to at most one, which preserves feasibility; the dual is compact because every edge is nonempty and hence each $y_E \leq 1$.

For completeness use the finitely generated closed cone

$$\mathcal{C} = \{(A^T z - s, \mathbf{1}^T z + t) : z, s, t \geq 0\} \subset \mathbb{R}^{m+1}.$$

For $\beta < \alpha$, the point $(\mathbf{1}, \beta)$ is outside it. Finite-dimensional separation yields (u, b) , nonnegative on this cone and strictly negative on that point. The generators $(-e_i, 0)$ and $(0, 1)$ give $u \leq 0$ and $b \geq 0$. With $y = -u$, vertex generators imply $Ay \leq b\mathbf{1}$, while strict separation gives $\sum_i y_i > b\beta$. If $b = 0$, nonempty edges and $y \geq 0$ would imply $y = 0$, a contradiction. Thus y/b is a feasible dual vector of value $> \beta$. Letting β increase to α and using dual compactness proves equality of optima.

The standard finite convex separation theorem is used here. Closedness of the finitely generated cone follows by choosing a representation with linearly independent generators, deleting dependencies while preserving nonnegative coefficients. Along a convergent sequence pass to one fixed independent subset of the finitely many generators; its coefficients then converge by the inverse on their linear span. This justifies the needed closedness.

References and provenance

[K] J. Kahn, *On a Problem of Erdős and Lovász. II: $n(r)=O(r)$* , J. Amer. Math. Soc. **7** (1994), 125–143. DOI <https://doi.org/10.1090/S0894-0347-1994-1224593-5>, Corollary D.5.4. The harmonic integer bound and rounding theorem are prior work.

[F] Repository note *A universal fractional cover envelope and its design equality cases*, commit 4ae9051, 7 October 2026. Its stronger noninteger envelope and exact equality classification are preserved. The present deletion estimate uses only the weighted-star inequality and supplies a separate quantitative stability theorem.

The limited literature screen also identifies Kayll’s DIMACS Report 95-55, Theorem D.2.1, as an existing resolution of Kahn’s Conjecture 5.6. That theorem is not an input here. The new finite extraction and fixed-ratio characterization remain subject to further novelty comparison; no first-result wording is used.

Part S: The complete matching spectrum

S.1. Definitions and statements

A hypergraph here is a finite simple family of nonempty finite sets. Its rank is at most an integer $r \geq 2$; its edge count is m . Its matching number $\nu(H)$ is the largest number of pairwise disjoint edges. Its fractional vertex-cover number $\tau^*(H)$ is the minimum sum of nonnegative vertex weights with sum at least one on an edge. Finite LP duality identifies this with the maximum $Y = \sum_E y_E$

under

$$y_E \geq 0, \quad \sum_{E \ni v} y_E \leq 1. \quad (\text{S.1})$$

The complete finite separation argument is given in Appendix S.A of [F]. Empty families have both numbers zero and are allowed below.

Recall the continuous nondecreasing function

$$\psi(x) = x/2 \quad (0 \leq x \leq 1), \quad \psi(x) = \max\{a/(a+1), x/(a+2)\} \quad (a \leq x \leq a+1, \ a = 1, 2, \dots). \quad (\text{S.2})$$

It is the sharp intersecting fractional frontier proved in [F]. For an integer $s \geq 1$, define

$$\Psi_s(c) = \max \left\{ \sum_{i=1}^s \psi(c_i) : c_i \geq 0, \sum_{i=1}^s c_i = c \right\}. \quad (\text{S.3})$$

This is an attained maximum on a compact simplex. For $s = 1$, it is exactly ψ .

Theorem S.1 (finite and sharp bounded-packing law). Every H with $\nu(H) \leq s$ satisfies

$$\tau^*(H) \leq r\Psi_s(m/r) + s/2. \quad (\text{S.4})$$

If s is fixed, $r_j \rightarrow \infty$, $\nu(H_j) \leq s$, and $m_j/r_j \rightarrow c < \infty$, then

$$\limsup_j \tau^*(H_j)/r_j \leq \Psi_s(c). \quad (\text{S.5})$$

For every real $c \geq 0$, a sequence of simple uniform hypergraphs of ranks tending to infinity attains equality in this limit and has matching number **exactly** s . The result imposes no partite, regularity, linearity or intersection-size assumption.

Here and below fixed-matching sharpness is an asymptotic statement, not an assertion that (S.4) is the best bound for every finite (r, m, s) .

Theorem S.2 (finite closed formula). Put $c = Q + \theta$, where $Q = \lfloor c \rfloor$ and $0 \leq \theta < 1$. For $s \geq 2$ set $u = s - 1$. For integers $0 \leq a \leq Q$, write $Q - a = ub + t$, with $0 \leq t < u$. Then

$$\Psi_s(c) = \max_{0 \leq a \leq Q} \left\{ \psi(a + \theta) + (u - t) \frac{b}{b+1} + t \frac{b+1}{b+2} \right\}. \quad (\text{S.6})$$

In particular, $\Psi_2(3) = 7/6$, $\Psi_2(7/2) = 7/6$, and $\Psi_2(15/4) = 5/4$. This formula involves a finite list, not a numerical optimization or an unverified solver assumption.

Define h by linear interpolation of its nonnegative integer values $h(a) = a/(a+1)$. Thus

$$h(x) = \frac{a}{a+1} + \frac{x-a}{(a+1)(a+2)} \quad (a \leq x \leq a+1). \quad (\text{S.7})$$

Theorem S.3 (arbitrary and diverging matching numbers). For every nonempty H , putting $s = \nu(H)$,

$$\frac{\tau^*(H)}{rs} \leq h\left(\frac{m}{rs}\right) + \frac{1}{2r}. \quad (\text{S.8})$$

For all fixed $c \geq 0$,

$$h(c) - \frac{1}{2s} \leq \frac{\Psi_s(sc)}{s} \leq h(c). \quad (\text{S.9})$$

Consequently when $r_j \rightarrow \infty$, $\nu(H_j) \rightarrow \infty$, and $m_j/(r_j\nu(H_j)) \rightarrow c < \infty$, the sharp upper limit of $\tau^*(H_j)/(r_j\nu(H_j))$ is $h(c)$. Uniform families attain this for any prescribed integer sequences of ranks and matching numbers tending to infinity.

S.2. The anchored finite inequality

The key point is that the earlier signed-bin proof does not require all pairs of edges in a group to intersect. It only requires an edge of maximum weight that meets every edge of that group.

Lemma S.4 (anchored bound). Let G have $t \geq 1$ edges of rank at most r , and let y be a vector satisfying (S.1) on G . Suppose $P \in G$, every other edge meets P , and $b = y_P \geq y_E$ for all E in G . Then, for every integer $k \geq 2$,

$$\sum_{E \in G} y_E \leq \max\{((k-1)r+1)/k, t/(k+1)\}. \quad (\text{S.10})$$

If $t \geq 2$, the same sum is at most $t/2$. If $t = 1$ it is at most one. In all cases,

$$\sum_{E \in G} y_E \leq r\psi(t/r) + 1/2. \quad (\text{S.11})$$

Proof. Write $Y_G = \sum_{E \in G} y_E$. The nonnegative incidence count on P gives $Y_G \leq r - (r-1)b$. If $b \leq 1/(k+1)$, use $Y_G \leq tb$; if $b > 1/k$, use that star bound. Both give (S.10). In the remaining interval $1/(k+1) < b \leq 1/k$, assign every other edge to exactly one of its intersection vertices with P . A bin of ℓ edges has total W with $W \leq \ell b$ and $W \leq 1 - b$. Put $B = 1 - kb \geq 0$ and $p = (k+1)b - 1 > 0$. For $\ell \leq k-1$,

$$W - B\ell \leq \ell p \leq (k-1)p;$$

for $\ell \geq k$,

$$W - B\ell \leq 1 - b - kB = (k-1)p.$$

Sum only over this genuine partition. Total mass is $Y_G - b$ and total count is $t-1$, so, with $g = (k-1)r + 1$,

$$Y_G \leq t - g + b\{(k+1)g - kt\}.$$

This is affine in b , taking the values $t/(k+1)$ and g/k at the two reciprocal endpoints. This proves (S.10), without any assumption about intersections between non-anchor edges.

For the half bound, every other edge satisfies $y_E + b \leq 1$. If $b \leq 1/2$, use $Y_G \leq tb \leq t/2$. Otherwise

$$Y_G \leq b + (t-1)(1-b) = t-1 - (t-2)b \leq t/2 \quad (t \geq 2).$$

For $t = 1$, nonemptiness of P gives $b \leq 1$. If $t \geq 2$ and $t/r \leq 1$, this half bound is exactly $r\psi(t/r)$. If $t/r \geq 1$, choose an integer $a \geq 1$ with $a \leq t/r \leq a+1$ and apply (S.10) with $k=a+1$. Its first term differs from $ra/(a+1)$ by $1/k \leq 1/2$, proving (S.11). For $t = 1$, $r\psi(1/r) + 1/2 = 1$; this also verifies (S.11).

S.3. Greedy partition and the finite law

Fix any feasible y on H . Among the remaining edges choose one of maximum weight, P_{-1} , and let G_{-1} consist of every remaining edge meeting P_{-1} , including P_{-1} itself. Remove G_{-1} . Repeat until no edge remains. At every step the selected edge meets all edges of its group and has maximum weight there. Its group's restricted vector is feasible.

All selected anchors are pairwise disjoint: a later selected edge was not removed as a neighbor of an earlier anchor. Hence the number l of groups is at most $\nu(H) \leq s$. Their edge sets partition H ;

write $t_i = |G_i|$, and pad the list to s groups with empty groups. Applying (S.11) to each nonempty group gives

$$Y \leq r \sum_{i=1}^s \psi(t_i/r) + l/2 \leq r\Psi_s(m/r) + s/2.$$

The exact partition identity $\sum_i t_i = m$ is what permits (S.3). This holds for every feasible vector, proving (S.4) by finite duality. Empty families cause no difficulty. The universal additive constant $s/2$ cannot be reduced: s disjoint r -edges have $m = s$, $\tau^* = s$, and $\Psi_s(s/r) = s/(2r)$, so (S.4) holds with exact equality.

Both ψ and Ψ_s are nondecreasing and $1/2$ -Lipschitz. For ψ , this follows from its piecewise slopes, all between zero and $1/2$. For Ψ_s , increasing total mass by δ can only increase the maximum: add δ to a coordinate of a maximizing allocation. Conversely trim δ from coordinates of any allocation at the larger total mass; the sum changes by at most $\delta/2$. Taking the maximum proves the Lipschitz bound. Dividing (S.4) by r and passing to the limit now proves (S.5).

S.4. Sharpness with a common rank

One must not silently combine unrelated rank subsequences. We prove the following common-rank construction explicitly.

Lemma S.5. For every fixed $x \geq 0$ and every sufficiently large integer R , there is a simple intersecting R -uniform family $H_R(x)$ with

$$|H_R(x)|/R \rightarrow x, \quad \tau^*(H_R(x))/R \rightarrow \psi(x). \quad (\text{S.12})$$

For fixed positive integer $x=a$ this uses only Wilson's named design existence theorem, not Kahn's covering corollary.

Proof. We use the exact constructions of [F], whose two published inputs are restated here. Wilson's theorem provides a Steiner 2 -($v, k, 1$) design for every sufficiently large admissible v with fixed k , subject to $k-1 \mid v-1$ and $k(k-1) \mid v(v-1)$. In particular choose $v = 1 + k(k-1)N$; replication is $r_0 = kN$. Its incidence dual has fractional value v/k , certified by uniform old-edge weights $1/k$ and old-vertex weights $1/r_0$. Kahn's 1994 Corollary S.5.4 states that an intersecting r_0 -uniform family with at most $C r_0$ edges, C fixed, and maximum pair intersection $o(r_0)$ has an integer cover of size $(C/(C+1) + o(1))r_0$. Here take $C=k$: the dual is linear and $v = (k-1)r_0 + 1 \leq k r_0$, so a cover C_0 smaller than r_0 exists. There are $v r_0/k = \Theta(r_0^2)$ old vertices. Enlarge C_0 to a set T of r_0-1 old vertices and add edges $T \cup \{p\}$ for distinct old p outside T , skipping original edges. There are $\Theta(r_0^2)$ candidates and only $O(r_0)$ forbidden originals, so any requested $O(r_0)$ additions are possible. Every added edge meets the originals through C_0 and other new edges through T . The old dual weights extended by zero and the old primal weights $1/r_0$ still agree at v/k . This proves the plateau extension explicitly.

If x lies on the plateau in $[a, a+1]$, take $k=a+1$, $N = \lfloor R/k \rfloor$, and $r_0 = R - O(1)$. Add $\max\{0, \lfloor xR \rfloor - v\}$ plateau edges. Privately pad all edges from rank r_0 to R . Put zero primal weight on every padding vertex. The old primal/dual certificates still agree at v/k , while the edge count divided by R tends to x . For $x=a$, only $O(1)$ additional edges could be requested, and taking no additional edges instead still yields (S.12); thus Kahn is unnecessary at integer x .

If $x > 0$ lies on a ramp, choose its block size $k=a+2$, or $k=2$ below one. Choose the largest admissible $v = 1 + k(k-1)N \leq xR$. Then $v = xR + O(1)$, and its replication $r_0 = (v-1)/(k-1) \leq R$ because $x \leq k-1$. Pad every edge privately to R . The same exact certificates give value v/k and (S.12). At

a switch either construction works. For $x=0$ take one R-edge. This completes the common-rank lemma. The prime-power alternatives in [F] give additional explicit subsequences; they are not used to assert construction at every large integer R.

Choose a maximizing allocation $(c_1, \dots, c_s)$ in (S.3). For each large R take the disjoint union of the s families $H_R(c_i)$, on disjoint vertex sets. It is simple and R-uniform. Each component is nonempty and intersecting, so its matching number is one; the union has matching number exactly s. Fractional cover is additive across disjoint components, by summing separate feasible primal and dual certificates. Formula (S.12) then attains (S.5). This proves Theorem S.1.

S.5. Reduction to a finite formula

On each integer unit interval ψ is convex: it is the maximum of a constant and an affine function, or the affine function $x/2$ below one. Take a maximizing allocation in (S.3). If two coordinates are noninteger, keep their sum and all other coordinates fixed, and move the two in opposite directions within their current closed unit intervals. The allowed parameter values form a nondegenerate closed segment. The sum of their two ψ values is convex on this segment, so one endpoint has value at least the current value. At an endpoint at least one of the two coordinates is integral. This replacement preserves a maximum and decreases the number of noninteger coordinates. After at most $s-1$ such replacements, all but at most one coordinate are integral. This elementary reduction uses only one-dimensional convexity, without an unproved enumeration of continuous allocations.

Thus a maximizing allocation can be chosen with all but at most one coordinate integral. For $c = Q + \theta$, the remaining coordinate is $a + \theta$, with integer $0 \leq a \leq Q$. The other $u=s-1$ integer coordinates sum to $Q-a$. Their integer values maximize $\sum f(n_i)$, where $f(n) = n/(n+1)$. The increment

$$f(n+1) - f(n) = 1/\{(n+1)(n+2)\}$$

is strictly decreasing. Transferring one unit from a coordinate at least two larger than another strictly increases the sum. The maximizing integer coordinates therefore differ by at most one: $u=t$ equal b and t equal $b+1$, where $Q-a = ub+t$. This proves (S.6), including $\theta = 0$. Zero integer coordinates contribute zero and are allowed. No unbounded search remains in that formula.

S.6. Concave limit and an exact gap distinction

The slopes in (S.7) decrease, so h is concave and $1/2$ -Lipschitz. Also $\psi \leq h$, with equality on $[0,1]$ and at integer points, and strict inequality at every noninteger point greater than one. To check this directly on $[a, a+1]$, h lies above its constant left endpoint and its chord to the right endpoint; each of the two terms in (S.2) touches it at only its respective integer endpoint. Jensen's inequality gives

$$\Psi_s(c) \leq sh(c/s). \quad (\text{S.13})$$

At integer total $c=Q$, balanced integer coordinates attain equality in (S.13), by the preceding unit-transfer argument. At arbitrary $c=Q+\theta$, increasing one coordinate in this integer allocation shows $\Psi_s(c) \geq sh(Q/s)$. Consequently

$$0 \leq sh(c/s) - \Psi_s(c) \leq (c-Q)/2 < 1/2. \quad (\text{S.14})$$

This proves (S.9), and (S.8) follows from (S.4) and (S.13). For $c \leq s$, choose all coordinates in $[0,1]$, giving $\Psi_s(c) = c/2$. For noninteger $c > s$ the inequality in (S.13) is strict: equality in Jensen requires all coordinates to lie in the same linear unit interval of h above one (or all equal at an integer

breakpoint). Equality $\psi = h$ then forces integral coordinates, whose total cannot be a noninteger. This explains why the fixed-s spectrum can be strictly smaller than the limiting concave curve.

Finally let the rank R and prescribed matching number s both tend to infinity, and write $c=a+\theta$ with integer $a \geq 0$. Use only the common-rank integer endpoint families $H_R(a)$ and $H_R(a+1)$, in proportions tending to $1-\theta$ and θ among s disjoint copies. Their edge density and fractional value, normalized by Rs , tend to c and $(1-\theta)f(a) + \theta f(a+1) = h(c)$. Their matching number is exactly s . Only two fixed design block sizes occur, so the common-rank error is $O(1/R)$, uniformly in the number of copies. This proves the sharpness and arbitrary prescribed-sequence clause in Theorem S.3. The case $c=0$ uses s disjoint R -edges.

S.7. Dependencies, verification and scope

The finite upper bound uses only feasible weights, the exact greedy partition and the anchored signed count. It uses neither Wilson nor Kahn nor any solver output. The common-rank lower constructions use explicitly named published inputs from [F]; the diverging-matching sharpness requires only Wilson’s design theorem at integer endpoints.

This is a fractional matching and cover result, not an integer-cover rounding theorem. It does not resolve Ryser, Kahn’s triple-intersection question or the general weighted nonuniform Füredi–Kahn–Seymour conjecture.

Pinned Lean coverage, exact rational diagnostics, source attribution and the frozen inventory are recorded in separate files. The full greedy construction, convex allocation reduction, imported existence theorems, LP duality and asymptotic sharpness are written proofs, not asserted to be whole-paper formalized. No external human review, historical priority or prize-level classification is claimed.

References

- [F] The repository proof *The sharp fractional cover frontier at every finite edge/rank ratio*, public commit <https://github.com/mxym/math/commit/c5255aaf66f9e894c6cbb0e2d1eb47c738c23765>, 7 October 2026. Its exact finite incidence Lean proof, construction certificates and full LP-duality appendix are preserved unchanged.
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- [K] J. Kahn, *On a Problem of Erdős and Lovász. II: $n(r)=O(r)$* , JAMS 7 (1994), 125–143, Corollary S.5.4, printed p.140, DOI <https://doi.org/10.1090/S0894-0347-1994-1224593-5>.
- [Fu] Z. Füredi, *Maximum degree and fractional matchings in uniform hypergraphs*, Combinatorica 1 (1981), 155–162. The classical rank/matching-number bound is prior context. The present spectrum also fixes the edge/rank ratio; no priority conclusion is drawn.
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